

Question	Answer	Mark	Mark scheme	Additional guidance
17	185	P1	for using the sine rule to find angle QSR , eg $\frac{26}{\sin 120} = \frac{12}{\sin QSR}$ OR for using the cosine rule to find SR eg $26^2 = 12^2 + SR^2 - 2 \times 12 \times SR \times \cos 120$ oe	Accept values to 3 figures or more, rounded or truncated throughout.
		P1	for complete process to find expression for $\sin QSR$, eg $\sin QSR = \frac{12 \times \sin 120}{26}$ (= 0.399....) or for $QSR = 23.55...$ OR for process to write rule as a quadratic equation in SR , eg $SR^2 + 12SR - 532$ (= 0)	
		P1	for process to find angle SQR , eg $180 - 120 - "23.5..."$ (= 36.44...) OR for process to find SR , eg substitution into the quadratic formula, $\frac{-12 \pm \sqrt{12^2 - 4 \times 1 \times -532}}{2 \times 1}$ oe (= 17.8...)	Do not award marks for answer if not supported by working.
		P1	for process to find area of triangle, eg $\frac{1}{2} \times 26 \times 12 \times \sin "36.4..."$ (= 92.66...) OR for process to find area of triangle, eg $\frac{1}{2} \times "17.83" \times 12 \times \sin 120$ (= 92.66...)	
		A1	for value in the range 185 to 186	